

>>> Topological invariant of 4-manifolds based on a 3-group

>>> *9th Tux Workshop on Quantum Gravity*

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>>> A sketch of the talk

► 3-group and 3-gauge theory

→ based on *R. Picken and J. Faria Martins*, *Diff. Geom. Appl.* 29, 179 (2011), [arXiv:0907.2566](#).

► $3BF$ action with constraints

→ Models with relevant dynamics *T. Radenković and M. Vojinović*, *J. High Energy Phys.* 10, 222 (2019), [arXiv:1904.07566](#).

► Quantization of the topological $3BF$ theory

→ The construction of the state sum Z and a proof that the $3BF$ state sum is invariant under Pachner moves.

T. Radenković and M. Vojinović, [arXiv: 2201.02572](#).

→ the state sum Z is an example of Porter's TQFT for $d = 4$ and $n = 3$
T. Porter, *J. Lond. Math. Soc.* (2)58, No. 3, 723 (1998), MR 1678163.

► Pachner move invariance – sketch of the proof

→ This is a generalization of the state sum based on the classical $2BF$ action with the underlying 2-group structure

F. Girelli, H. Pfeiffer and E. M. Popescu, *Jour. Math. Phys.* 49, 032503 (2008), [arXiv:0708.3051](#).

► Conclusions

>>> 3-groups

2-crossed module $(L \xrightarrow{\delta} H \xrightarrow{\partial} G, \triangleright, \{_, _\}_P)$

- * Groups G , H , and L ;
- * maps ∂ and δ ($\partial\delta = 1_G$);
- * an action $\triangleright$ of the group G on all three groups;
- * a map $\{_, _\}_P$ called the *Peiffer lifting*:

$$\{_, _\}_P : H \times H \rightarrow L.$$

Certain axioms hold true among all these maps:

1. $\delta(\{h_1, h_2\}_P) = \langle h_1, h_2 \rangle_P, \quad \forall h_1, h_2 \in H,$
2. $[l_1, l_2] = \{\delta(l_1), \delta(l_2)\}_P, \quad \forall l_1, l_2 \in L.$ Here, the notation $[l, k] = lkl^{-1}k^{-1}$ is used;
3. $\{h_1 h_2, h_3\}_P = \{h_1, h_2 h_3 h_2^{-1}\}_P \partial(h_1) \triangleright \{h_2, h_3\}_P, \quad \forall h_1, h_2, h_3 \in H;$
4. $\{h_1, h_2 h_3\}_P = \{h_1, h_2\}_P \{h_1, h_3\}_P \{\langle h_1, h_3 \rangle_P^{-1}, \partial(h_1) \triangleright h_2\}_P, \quad \forall h_1, h_2, h_3 \in H;$
5. $\{\delta(l), h\}_P \{h, \delta(l)\}_P = l(\partial(h) \triangleright l^{-1}), \quad \forall h \in H, \quad \forall l \in L.$

>>> The **3BF** theory

One can now generalize the notion of parallel transport from curves to surfaces and volumes.

- * Given a 2-crossed module, one can define a 3-connection, an ordered triple (α, β, γ) , where α , β , and γ are algebra-valued differential forms,

$$\begin{aligned}\alpha &= \alpha^\alpha{}_\mu \tau_\alpha dx^\mu, & \alpha &\in \mathcal{A}^1(\mathcal{M}_4, \mathfrak{g}), \\ \beta &= \beta^a{}_{\mu\nu} t_a dx^\mu \wedge dx^\nu, & \beta &\in \mathcal{A}^2(\mathcal{M}_4, \mathfrak{h}), \\ \gamma &= \gamma^A{}_{\mu\nu\rho} T_A dx^\mu \wedge dx^\nu \wedge dx^\rho, & \gamma &\in \mathcal{A}^3(\mathcal{M}_4, \mathfrak{l}).\end{aligned}\tag{1}$$

- * Then introduce the line, surface and volume holonomies,

$$g = \mathcal{P} \exp \int_\gamma \alpha, \quad h = \mathcal{S} \exp \int_S \beta, \quad l = \mathcal{V} \exp \int_V \gamma.\tag{2}$$

- * The corresponding fake 3-curvature $(\mathcal{F}, \mathcal{G}, \mathcal{H})$ is defined as:

$$\begin{aligned}\mathcal{F} &= d\alpha + \alpha \wedge \alpha - \partial\beta, & \mathcal{G} &= d\beta + \alpha \wedge^\triangleright \beta - \delta\gamma, \\ \mathcal{H} &= d\gamma + \alpha \wedge^\triangleright \gamma + \{\beta \wedge \beta\}_{\text{pf}}.\end{aligned}\tag{3}$$

>>> The **3BF** theory

At this point one can construct the so-called **3BF** theory.

- * For a manifold $\mathcal{M}_4$ and the 2-crossed module $(L \xrightarrow{\delta} H \xrightarrow{\partial} G, \triangleright, \{_, _\}_{\text{pf}})$, that gives rise to 3-curvature $(\mathcal{F}, \mathcal{G}, \mathcal{H})$, one defines the **3BF** action as

$$S_{3BF} = \int_{\mathcal{M}_4} \langle B \wedge \mathcal{F} \rangle_{\mathfrak{g}} + \langle C \wedge \mathcal{G} \rangle_{\mathfrak{h}} + \langle D \wedge \mathcal{H} \rangle_{\mathfrak{l}}. \quad (4)$$

- * **3BF** theory is a topological gauge theory,
- * it is based on the 3-group structure,
- * it is a generalization of an ordinary **BF** theory for a given Lie group G .
- * The physical interpretation of the Lagrange multipliers C and D :
 - * the $\mathfrak{h}$ -valued 1-form C can be interpreted as the tetrad field if $H = \mathbb{R}^4$ is the spacetime translation group,

$$C \rightarrow e = e^a{}_{\mu}(x) t_a dx^{\mu}, \quad (5)$$

- * the $\mathfrak{l}$ -valued 0-form D can be interpreted as the set of real-valued matter fields, given some Lie group L :

$$D \rightarrow \phi = \phi^A(x) T_A. \quad (6)$$

>>> Constrained $3BF$ action

- * For a manifold $\mathcal{M}_4$ and the 2-crossed module $(L \xrightarrow{\delta} H \xrightarrow{\partial} G, \triangleright, \{_, _\}_{\text{pf}})$, that gives rise to 3-curvature $(\mathcal{F}, \mathcal{G}, \mathcal{H})$, one defines the $3BF$ action as

$$S_{3BF} = \int_{\mathcal{M}_4} \langle B \wedge \mathcal{F} \rangle_{\mathfrak{g}} + \langle C \wedge \mathcal{G} \rangle_{\mathfrak{h}} + \langle D \wedge \mathcal{H} \rangle_{\mathfrak{l}}. \quad (7)$$

- * Physically relevant models - The constrained $2BF$ actions describing the *Yang-Mills field* and *Einstein-Cartan gravity*, and constrained $3BF$ actions describing the *Klein-Gordon*, *Dirac*, *Weyl* and *Majorana fields* coupled to Yang-Mills fields and gravity in the standard way are formulated.

→ *T. Radenković and M. Vojinović*, J. High Energy Phys.10, 222 (2019), [arXiv:1904.07566](https://arxiv.org/abs/1904.07566).

>>> 3-gauge theory

- * Curves are labeled with the elements of G , and the elements are composed as

$$\bullet \xleftarrow{g_1} \bullet \xleftarrow{g_2} \bullet = \bullet \xleftarrow{g_1 g_2} \bullet$$

- * Surfaces are labeled with the elements $h \in H$. We split the boundary into two curves, the source curve $g_1 \in G$ and the target curve $g_2 \in G$,

$$\bullet \begin{array}{c} \xleftarrow{g_1} \\ \Downarrow h \\ \xleftarrow{g_2} \end{array} \bullet ,$$

so that the surface $h \in H$ satisfies:

$$\partial(h) = g_2 g_1^{-1} .$$

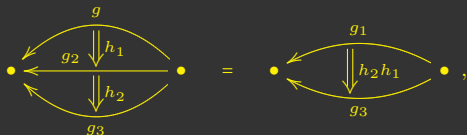
- * Volumes are labeled with the elements $l \in L$. We split the boundary into the source surface $\partial_3^-(l) = h_1$ and the target surface $\partial_3^+(l) = h_2$, and the common boundary of h_1 and h_2 we split into the source curve $\partial_2^-(l) = g_1$ and the target curve $\partial_2^+(l) = g_2$,

$$\bullet \begin{array}{c} \xleftarrow{g_1} \\ \Downarrow h_1 \\ \xleftarrow{g_2} \end{array} \bullet \quad \xRightarrow{l} \quad \bullet \begin{array}{c} \xleftarrow{g_1} \\ \Downarrow h_2 \\ \xleftarrow{g_2} \end{array} \bullet ,$$

$$\delta(l) = h_2 h_1^{-1} .$$

>>> 3-gauge theory

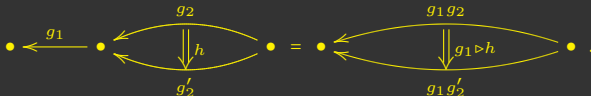
- * *Vertical composition of 2-morphisms.* One can compose 2-morphisms (g_1, h_1) and (g_2, h_2) vertically, when they are compatible, when $\partial_2^+(h_1) = \partial_2^-(h_2)$,



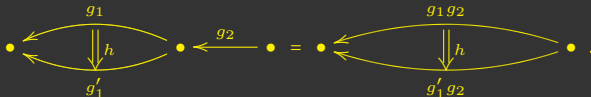
results in a 2-morphism $(g_1, h_2 h_1)$,

$$(g_2, h_2) \#_2 (g_1, h_1) = (g_1, h_2 h_1). \quad (8)$$

- * *Whiskering.* One can whisker a 2-morphism h with a morphism g_1 by attaching the whisker g_1 to the surface h from the left, such that $\partial_1^-(g_1) = \partial_1^+(h)$,

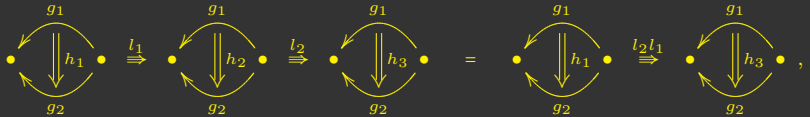


One can whisker g_2 to a surface h from the right, such that $\partial_1^-(h) = \partial_1^+(g_2)$,



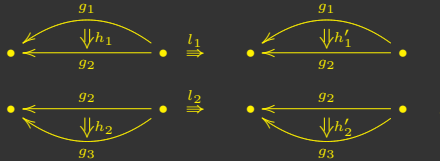
>>> 3-gauge theory

- * *Upward composition*. The upward composition of 3-morphisms (g_1, h_1, l_1) and (g_1, h_2, l_2) , when they are compatible, when $\partial_3^+(l_1) = \partial_3^-(l_2)$,

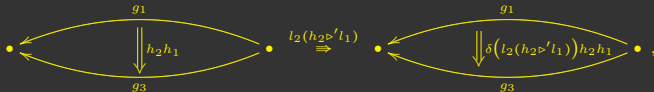


$$(g_1, h_2, l_2) \#_3 (g_1, h_1, l_1) = (g_1, h_1, l_2 l_1). \quad (9)$$

- * *Vertical composition*. The vertical composition of two 3-morphisms (g_1, h_1, l_1) and (g_2, h_2, l_2) , when they are compatible, when $\partial_2^+(l_1) = \partial_2^-(l_2)$,



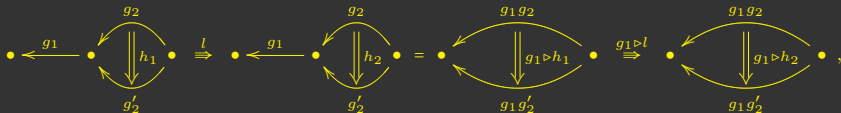
results in a 3-morphism $(g_1, h_2 h_1, l_2 (h_2 \triangleright' l_1))$,



$$(g_2, h_2, l_2) \#_2 (g_1, h_1, l_1) = (g_1, h_2 h_1, l_2 (h_2 \triangleright' l_1)). \quad (10)$$

>>> 3-gauge theory

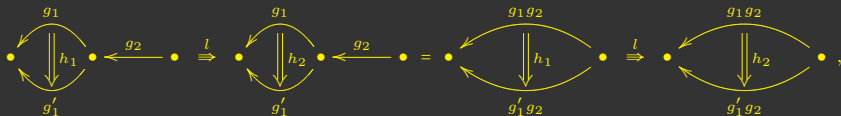
- * *Whiskering of the 3-morphisms with morphisms.* Whiskering of a 3-morphism by a morphism from the left is the composition of a volume $l \in L$ and curve $g_1 \in G$ from the left, when they are compatible, when $\partial_1^+(l) = \partial_1^-(g_1)$,



$$\begin{array}{c}
 \bullet \xleftarrow{g_1} \bullet \begin{array}{c} \curvearrowright^{g_2} \\ \Downarrow h_1 \\ \curvearrowleft_{g'_2} \end{array} \bullet \xrightarrow{l} \bullet \xleftarrow{g_1} \bullet \begin{array}{c} \curvearrowright^{g_2} \\ \Downarrow h_2 \\ \curvearrowleft_{g'_2} \end{array} \bullet = \bullet \begin{array}{c} \curvearrowright^{g_1 g_2} \\ \Downarrow g_1 \triangleright h_1 \\ \curvearrowleft_{g_1 g'_2} \end{array} \bullet \xrightarrow{g_1 \triangleright l} \bullet \begin{array}{c} \curvearrowright^{g_1 g_2} \\ \Downarrow g_1 \triangleright h_2 \\ \curvearrowleft_{g_1 g'_2} \end{array} \bullet ,
 \end{array}$$

$$g_1 \#_1 (g_2, h_1, l) = (g_1 g_2, g_1 \triangleright h, g_1 \triangleright l) . \quad (11)$$

One can whisker a 3-morphism by a morphism from the right, when they are compatible, $\partial_1^-(l) = \partial_1^+(g_2)$,

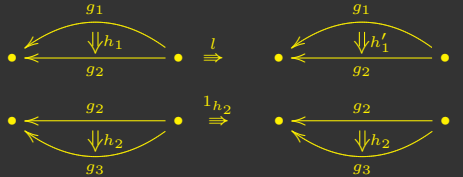


$$\begin{array}{c}
 \bullet \begin{array}{c} \curvearrowright^{g_1} \\ \Downarrow h_1 \\ \curvearrowleft_{g'_1} \end{array} \bullet \xleftarrow{g_2} \bullet \xrightarrow{l} \bullet \begin{array}{c} \curvearrowright^{g_1} \\ \Downarrow h_2 \\ \curvearrowleft_{g'_1} \end{array} \bullet \xleftarrow{g_2} \bullet = \bullet \begin{array}{c} \curvearrowright^{g_1 g_2} \\ \Downarrow h_1 \\ \curvearrowleft_{g'_1 g_2} \end{array} \bullet \xrightarrow{l} \bullet \begin{array}{c} \curvearrowright^{g_1 g_2} \\ \Downarrow h_2 \\ \curvearrowleft_{g'_1 g_2} \end{array} \bullet ,
 \end{array}$$

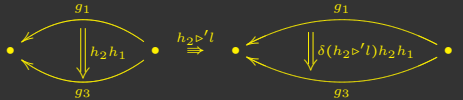
$$(g_1, h_1, l) \#_{1g_2} = (g_1 g_2, h_1, l) . \quad (12)$$

>>> 3-gauge theory

- * *Whiskering of 3-morphisms with 2-morphisms.* Whiskering of a 3-morphism with a 2-morphisms from below, when they are compatible, $\partial_2^+(l) = \partial_2^-(h_2)$, is formed as a vertical composition of 3-morphisms (g_1, h_1, l) and (g_2, h_2, l_{h_2}) ,



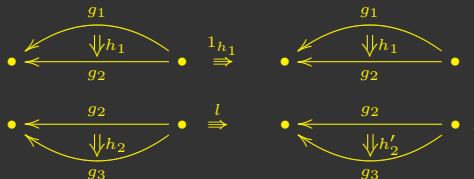
which results in a 3-morphism



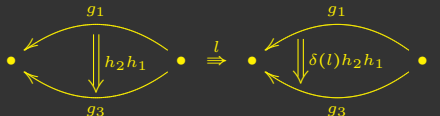
$$(g_1, h_1, l) \#_2 (g_2, h_2) = (g_1, h_2h_1, h_2 \triangleright' l) . \quad (13)$$

>>> 3-gauge theory

- * Whiskering a 3-morphism by 2-morphism from above, when they are compatible, when $\partial_2^-(l) = \partial_2^+(h_1)$,



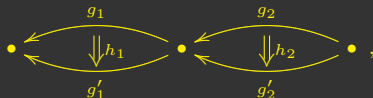
results in a 3-morphism,



$$(g_1, h_1) \#_2 (g_2, h_2, l) = (g_1, h_2 h_1, l). \quad (14)$$

>>> 3-gauge theory

- * *The interchanging 3-arrow.* The horizontal composition of two 2-morphisms h_1 and h_2 , when they are compatible, when $\partial_1^-(h_1) = \partial_1^+(h_2)$,



that results in a 3-morphism l , with source surface

$$\partial_3^-(l) = ((g_1, h_1) \#_1 g'_2) \#_2 (g_1 \#_1 (g_2, h_2)),$$

and target surface

$$\partial_3^+(l) = (g'_1 \#_1 (g_2, h_2)) \#_2 ((g_1, h_1) \#_1 g_2),$$



One obtains,

$$(g_1, h_1) \#_1 (g_2, h_2) = (g_1 g_2, h_1 g_1 \triangleright h_2, l), \quad (15)$$

where the 3-morphism l is Peiffer lifting $\{h_1, g_1 \triangleright h_2\}_p^{-1}$.

>>> 3-gauge theory

Lemma (1)

Let us consider a triangle, (jkl) . The edges $(jk), j < k$, are labeled by group elements $g_{jk} \in G$ and the triangle $(jkl), j < k < \ell$, by element $h_{jkl} \in H$.

$$\begin{array}{c}
 \begin{array}{ccc}
 l \bullet & k \bullet & j \bullet \\
 \swarrow g_{kl} & \nwarrow g_{jk} & \\
 & \Downarrow h_{jkl} & \\
 & g_{jl} &
 \end{array}
 =
 \begin{array}{ccc}
 l \bullet & k \bullet & j \bullet \\
 \swarrow 1_{\bullet} & \nwarrow g_{kl} & \\
 \Downarrow h_{jkl} & & \\
 \partial(h_{jkl}) & & \\
 \swarrow g_{kl} & \nwarrow g_{jk} & \\
 & \Downarrow 1_{g_{kl}g_{jk}} & \\
 & g_{kl}g_{jk} &
 \end{array}
 =
 \begin{array}{ccc}
 l \bullet & k \bullet & j \bullet \\
 \swarrow g_{kl} & \nwarrow g_{jk} & \\
 & \Downarrow h_{jkl} & \\
 & \partial(h_{jkl})g_{kl}g_{jk} &
 \end{array}
 \end{array}
 \quad (16)$$

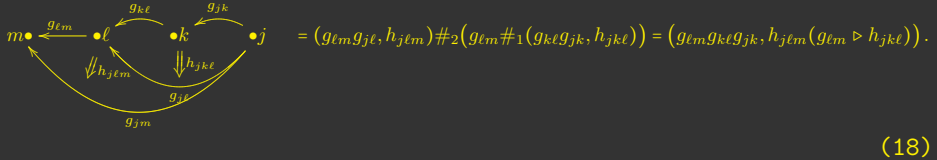
The curve $\gamma_1 = g_{kl}g_{jk}$ is the source and the curve $\gamma_2 = g_{jl}$ is the target of the surface morphism $\Sigma: \gamma_1 \rightarrow \gamma_2$, labeled by the group element h_{jkl} ,

$$g_{jl} = \partial(h_{jkl})g_{kl}g_{jk} . \quad (17)$$

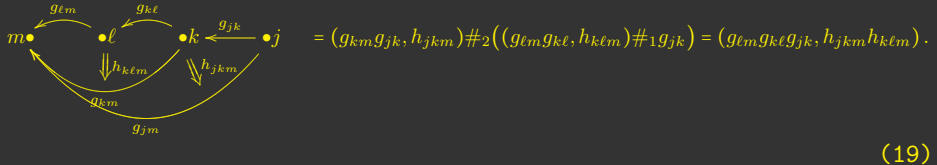
>>> 3-gauge theory

Lemma (2)

Let us consider a tetrahedron, $(jklm)$.



$$\begin{aligned}
 & \text{Diagram (18)} = (g_{lm} g_{jl}, h_{jlm}) \#_2 (g_{lm} \#_1 (g_{kl} g_{jk}, h_{jkl})) = (g_{lm} g_{kl} g_{jk}, h_{jlm} (g_{lm} \triangleright h_{jkl})).
 \end{aligned}
 \tag{18}$$



$$\begin{aligned}
 & \text{Diagram (19)} = (g_{km} g_{jk}, h_{jkm}) \#_2 ((g_{lm} g_{kl}, h_{klm}) \#_1 g_{jk}) = (g_{lm} g_{kl} g_{jk}, h_{jkm} h_{klm}).
 \end{aligned}
 \tag{19}$$

Moving from surface shown on the diagram (18) to the surface shown on the diagram (19) is determined by the group element l_{jklm} ,

$$h_{jkm} h_{klm} = \delta(l_{jklm}) h_{jlm} (g_{lm} \triangleright h_{jkl}).
 \tag{20}$$

>>> 3-gauge theory

Lemma (3)

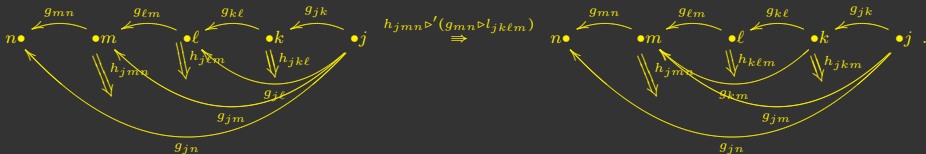
We consider a 4-simplex, $(jklmn)$. We cut the 4-simplex volume along the surface $h_{jmn}g_{mn} \triangleright (h_{jlm}g_{lm} \triangleright h_{jkl})$.

Step 1.

- * We move the surface from $h_{jlm}g_{lm} \triangleright h_{jkl}$ to $h_{jkm}h_{k\ell m}$ with the 3-arrow l_{jklm} .
- * To compose the resulting 3-morphism with surface h_{jmn} one must first whisker it from the left with g_{mn} .
- * The obtained 3-morphism $(g_{mn}g_{lm}g_{kl}g_{jk}, g_{mn} \triangleright (h_{jlm}g_{lm} \triangleright h_{jkl}), g_{mn} \triangleright l_{jklm})$ can be whiskered from below with the 2-morphism $(g_{mn}g_{jm}, h_{jmn})$.
- * The resulting 3-morphism is

$$(g_{mn}g_{lm}g_{kl}g_{jk}, h_{jmn}g_{mn} \triangleright (h_{jlm}g_{lm} \triangleright h_{jkl}), h_{jmn} \triangleright' (g_{mn} \triangleright l_{jklm}))$$

$$\Sigma_1 \rightarrow \Sigma_2, \quad \Sigma_1 = h_{jmn}g_{mn} \triangleright (h_{jlm}g_{lm} \triangleright h_{jkl}) \quad \text{and} \quad \Sigma_2 = h_{jmn}g_{mn} \triangleright (h_{jkm}h_{k\ell m}).$$



>>> 3-gauge theory

Lemma (3)

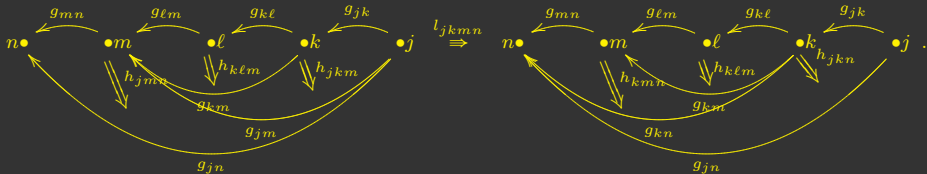
Step 2.

Let us move the surface to $h_{jkn}h_{kmn}g_{ml} \triangleright h_{k\ell m}$.

- * We consider the 3-morphism $(g_{mn}g_{km}g_{jk}, h_{jmn}g_{mn} \triangleright h_{jkm}, l_{jkmn})$ with the source surface $h_{jmn}g_{mn} \triangleright h_{jkm}$ and target surface $h_{jkn}h_{kmn}$.
- * This 3-morphism can be whiskered from above with the 2-morphism $(g_{mn}g_{\ell\ell}g_{k\ell}g_{jk}, g_{mn} \triangleright h_{k\ell m})$.
- * The obtained 3-morphism is

$$(g_{mn}g_{\ell m}g_{k\ell}g_{jk}, h_{jmn}g_{mn} \triangleright (h_{jkm}h_{k\ell m}), l_{jkmn})$$

$\Sigma_1 \rightarrow \Sigma_2$, $\Sigma_1 = h_{jmn}g_{mn} \triangleright (h_{jkm}h_{k\ell m})$ and $\Sigma_2 = h_{jkn}h_{kmn}g_{mn} \triangleright h_{k\ell m}$.



(22)

>>> 3-gauge theory

Lemma (3)

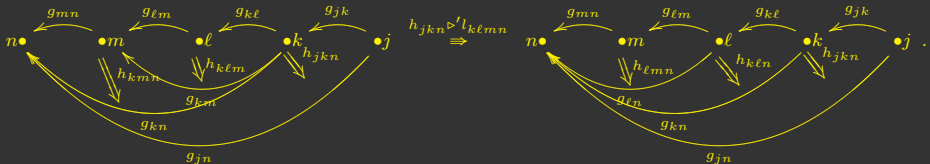
Step 3.

Next, we want to move the surface $h_{jkn}h_{kmn}g_{mn} \triangleright h_{k\ell m}$ to surface $h_{jkn}h_{k\ell n}h_{\ell mn}$.

- * We whisker the 3-morphism $(g_{mn}g_{\ell m}g_{k\ell}, h_{kmn}g_{mn} \triangleright h_{k\ell m}, l_{k\ell mn})$, with the source surface $h_{kmn}g_{mn} \triangleright h_{k\ell m}$ and target surface $h_{k\ell n}h_{\ell mn}$, with the morphism g_{jk} from the right.
- * The obtained the 3-morphism $(g_{mn}g_{\ell m}g_{k\ell}g_{jk}, h_{kmn}g_{mn} \triangleright h_{k\ell m}, l_{k\ell mn})$ we whisker with the 2-morphism $(g_{kn}g_{jk}, h_{jkn})$ from below.
- * We obtain the 3-morphism

$$(g_{mn}g_{\ell m}g_{k\ell}g_{jk}, h_{jkn}h_{kmn}g_{mn} \triangleright h_{k\ell m}, h_{jkn} \triangleright' l_{k\ell mn})$$

$\Sigma_1 \rightarrow \Sigma_2$, $\Sigma_1 = h_{jkn}h_{kmn}g_{mn} \triangleright h_{k\ell m}$ and $\Sigma_2 = h_{jkn}h_{k\ell n}h_{\ell mn}$.



(23)

>>> 3-gauge theory

Lemma (3)

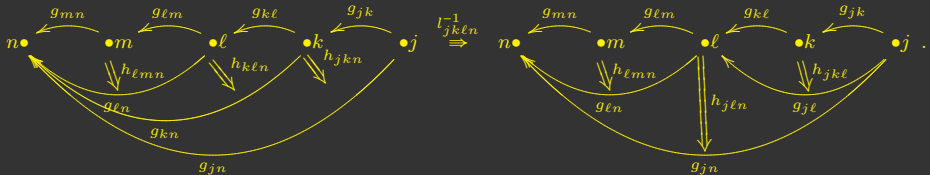
Step 4.

We map the surface $h_{jkn}h_{k\ell n}h_{\ell mn}$ to the surface $h_{j\ell n}g_{\ell n} \triangleright h_{jkl}h_{\ell mn}$.

- * The 3-morphism with the appropriate source and target is constructed by whiskering the 3-morphism $(g_{\ell n}g_{kl}g_{jk}, h_{jkn}h_{k\ell n}, l_{jkl\ell n}^{-1})$ with 2-morphism $(g_{mn}g_{\ell m}g_{kl}g_{jk}, h_{\ell mn})$ from above.
- * The obtained 3-morphism is

$$(g_{mn}g_{\ell m}g_{kl}g_{jk}, h_{jkn}h_{k\ell n}h_{\ell mn}, l_{jkl\ell n}^{-1})$$

$$\Sigma_1 \rightarrow \Sigma_2, \quad \Sigma_1 = h_{jkn}h_{k\ell n}h_{\ell mn} \quad \text{and} \quad \Sigma_2 = h_{j\ell n}g_{\ell n} \triangleright h_{jkl}h_{\ell mn}.$$



(24)

>>> 3-gauge theory

Lemma (3)

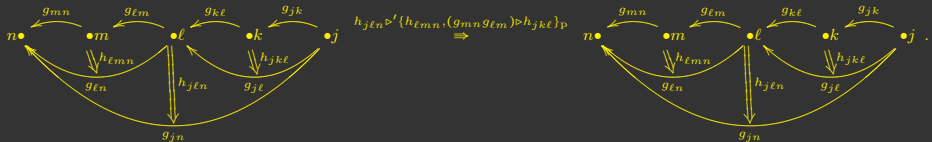
Step 5.

Next we map the surface $h_{j\ell n}g_{\ell n} \triangleright h_{jkl}h_{\ell mn}$ to the surface $h_{j\ell n}h_{\ell mn}(g_{mn}g_{\ell m}) \triangleright h_{jkl}$.

- * We use the inverse interchanging 2-arrow composition to map the surface $g_{\ell n} \triangleright h_{jkl}h_{\ell mn}$ to the surface $h_{\ell mn}(g_{mn}g_{\ell m}) \triangleright h_{jkl}$, resulting in the 3-morphism $(g_{mn}g_{\ell m}g_{kl}g_{jk}, g_{\ell n} \triangleright h_{jkl}h_{\ell mn}, \{h_{\ell mn}, (g_{mn}g_{\ell m}) \triangleright h_{jkl}\}_P)$.
- * Next, we whisker the obtained 3-morphism with the 2-morphism $(g_{\ell n}g_{j\ell}, h_{j\ell n})$ from below.
- * The obtained 3-morphism with the appropriate source and target surfaces is

$$(g_{mn}g_{\ell m}g_{kl}g_{jk}, h_{j\ell n}g_{\ell n} \triangleright h_{jkl}h_{\ell mn}, h_{j\ell n} \triangleright' \{h_{\ell mn}, (g_{mn}g_{\ell m}) \triangleright h_{jkl}\}_P)$$

$\Sigma_1 \rightarrow \Sigma_2$, $\Sigma_1 = h_{j\ell n}g_{\ell n} \triangleright h_{jkl}h_{\ell mn}$ and $\Sigma_2 = h_{j\ell n}h_{\ell mn}(g_{mn}g_{\ell m}) \triangleright h_{jkl}$.



(25)

>>> 3-gauge theory

Lemma (3)

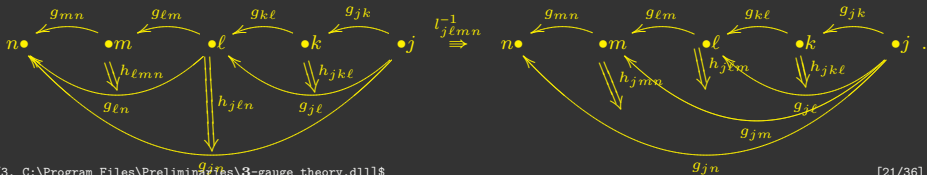
Step 6.

Finally, we construct the 3-morphism that maps the surface $h_{j\ell n}h_{\ell mn}(g_{mn}g_{\ell m}) \triangleright h_{jkl}$ to the starting surface $h_{jmn}g_{mn} \triangleright (h_{j\ell m}g_{\ell m} \triangleright h_{jkl})$.

- * To obtain the 3-morphism with the appropriate source and target surfaces we first move the surface $h_{j\ell n}h_{\ell mn}$ to the surface $h_{jmn}g_{mn} \triangleright h_{j\ell m}$ with the 3-arrow $(g_{mn}g_{\ell m}g_{j\ell}, h_{j\ell n}h_{\ell mn}, l_{j\ell mn}^{-1})$.
- * Next, we whisker the 3-morphism $(g_{mn}g_{\ell m}g_{j\ell}, h_{j\ell n}h_{\ell mn}, l_{j\ell mn}^{-1})$ with the 2-morphism $(g_{mn}g_{\ell m}g_{kl}g_{jk}, (g_{mn}g_{\ell m}) \triangleright h_{jkl})$ from above.
- * The obtained 3-morphism

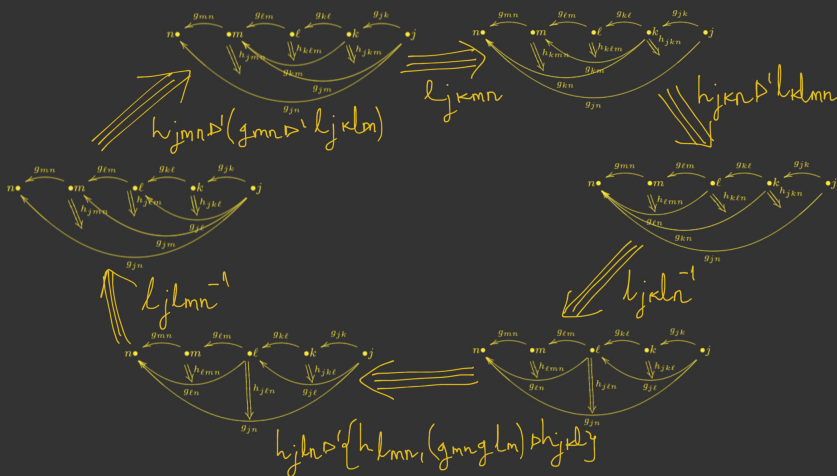
$$(g_{mn}g_{\ell m}g_{kl}g_{jk}, h_{j\ell n}h_{\ell mn}(g_{mn}g_{\ell m}) \triangleright h_{jkl}, l_{j\ell mn}^{-1})$$

$\Sigma_1 \rightarrow \Sigma_2$, $\Sigma_1 = h_{j\ell n}h_{\ell mn}(g_{mn}g_{\ell m}) \triangleright h_{jkl}$ and $\Sigma_2 = h_{jmn}g_{mn} \triangleright (h_{j\ell m}g_{\ell m} \triangleright h_{jkl})$.



>>> 3-gauge theory

Lemma (3)



>>> 3-gauge theory

Lemma (3)

After the upward composition of the 3-morphisms given by the diagrams (21)-(26), the obtained 3-morphism is:

$$\begin{aligned} & (g_{mn}g_{\ell m}g_{kl}g_{jk}, h_{j\ell n}h_{\ell mn}(g_{mn}g_{\ell m}) \triangleright h_{jkl}, l_{j\ell mn}^{-1})\#_3 \\ & (g_{mn}g_{\ell m}g_{kl}g_{jk}, g_{\ell n} \triangleright h_{jkl}h_{\ell mn}, h_{j\ell n} \triangleright' \{h_{\ell mn}, (g_{mn}g_{\ell m}) \triangleright h_{jkl}\}_p)\#_3 \\ & (g_{mn}g_{\ell m}g_{kl}g_{jk}, h_{jkn}h_{k\ell n}h_{\ell mn}, l_{j\ell n}^{-1})\#_3 \\ & (g_{mn}g_{\ell m}g_{kl}g_{jk}, h_{jkn}h_{kmn}g_{ml} \triangleright h_{k\ell m}, h_{jkn} \triangleright' l_{jkmn})\#_3 \\ & (g_{mn}g_{\ell m}g_{kl}g_{jk}, h_{jmn}g_{mn} \triangleright (h_{jkm}h_{k\ell m}), l_{jkmn})\#_3 \\ & (g_{mn}g_{\ell m}g_{kl}g_{jk}, h_{jmn}g_{mn} \triangleright (h_{j\ell m}g_{\ell m} \triangleright h_{jkl}), h_{jmn} \triangleright' (g_{mn} \triangleright l_{jk\ell m})) \\ & = (g_{mn}g_{\ell m}g_{kl}g_{jk}, h_{jmn}g_{mn} \triangleright (h_{j\ell m}g_{\ell m} \triangleright h_{jkl}), l_{j\ell mn}^{-1} h_{j\ell n} \triangleright' \{h_{\ell mn}, (g_{mn}g_{\ell m}) \triangleright h_{jkl}\}_p \\ & \quad l_{j\ell n}^{-1} (h_{jkn} \triangleright' l_{k\ell mn}) l_{jkmn} h_{jmn} \triangleright' (g_{mn} \triangleright l_{jk\ell m})). \end{aligned} \tag{27}$$

The obtained 3-morphism is the identity morphism with source and target surface $\mathcal{V}_1 = \mathcal{V}_2 = h_{jmn}g_{mn} \triangleright (h_{j\ell m}g_{\ell m} \triangleright h_{jkl})$,

$$l_{j\ell mn}^{-1} h_{j\ell n} \triangleright' \{h_{\ell mn}, (g_{mn}g_{\ell m}) \triangleright h_{jkl}\}_p l_{j\ell n}^{-1} (h_{jkn} \triangleright' l_{k\ell mn}) l_{jkmn} h_{jmn} \triangleright' (g_{mn} \triangleright l_{jk\ell m}) = e. \tag{28}$$

>>> Quantization of the topological **3BF** theory

We want to construct a *state sum model* from the classical S_{3BF} action by the usual spinfoam quantization procedure.

$$Z = \int \mathcal{D}\alpha \mathcal{D}\beta \mathcal{D}\gamma \mathcal{D}B \mathcal{D}C \mathcal{D}D \exp \left(i \int_{M_4} \langle B \wedge \mathcal{F} \rangle_{\mathfrak{g}} + \langle C \wedge \mathcal{G} \rangle_{\mathfrak{h}} + \langle D \wedge \mathcal{H} \rangle_{\mathfrak{l}} \right). \quad (29)$$

↪ The formal integration over the Lagrange multipliers B , C , and D leads to:

$$Z = \mathcal{N} \int \mathcal{D}\alpha \mathcal{D}\beta \mathcal{D}\gamma \delta(\mathcal{F})\delta(\mathcal{G})\delta(\mathcal{H}). \quad (30)$$

↪ Discretization of the 3-connection:

- ▶ $\alpha \in \mathcal{A}^1(\mathcal{M}_4, \mathfrak{g}) \mapsto g_{\epsilon} \in G$ coloring the edges $\epsilon = (jk) \in \Lambda_1$,
- ▶ $\beta \in \mathcal{A}^2(\mathcal{M}_4, \mathfrak{h}) \mapsto h_{\Delta} \in H$ coloring the triangles $\Delta = (jkl) \in \Lambda_2$,
- ▶ $\gamma \in \mathcal{A}^3(\mathcal{M}_4, \mathfrak{l}) \mapsto l_{\tau} \in L$ coloring the tetrahedrons $\tau = (jklm) \in \Lambda_3$.

$$\left. \begin{array}{ll} \int \mathcal{D}\alpha & \mapsto \Pi_{(jk) \in \Lambda_1} \int_G dg_{jk} \\ \int \mathcal{D}\beta & \mapsto \Pi_{(jkl) \in \Lambda_2} \int_H dh_{jkl} \\ \int \mathcal{D}\gamma & \mapsto \Pi_{(jklm) \in \Lambda_3} \int_L dl_{jklm} \end{array} \right\} \longrightarrow \text{The discretization of path integral measures.}$$

>>> Quantization of the etiological $3BF$ theory

↪ The condition $\delta(\mathcal{F})$ is discretized as

$$\delta(\mathcal{F}) = \prod_{(jk\ell) \in \Lambda_2} \delta_G(g_{jk\ell}), \quad \delta_G(g_{jk\ell}) = \delta_G(\partial(h_{jk\ell}) g_{k\ell} g_{jk} g_{j\ell}^{-1}). \quad (31)$$

↪ The condition $\delta(\mathcal{G})$ on the fake curvature 3-form reads

$$\delta(\mathcal{G}) = \prod_{(jk\ell m) \in \Lambda_3} \delta_H(h_{jk\ell m}), \quad (32)$$

$$\delta_H(h_{jk\ell m}) = \delta_H(\delta(l_{jk\ell m}) h_{j\ell m} (g_{\ell m} \triangleright h_{jk\ell}) h_{k\ell m}^{-1} h_{jkm}^{-1}). \quad (33)$$

↪ The condition $\delta(\mathcal{H})$ is discretized as

$$\delta(\mathcal{H}) = \prod_{(jk\ell mn) \in \Lambda_4} \delta_L(l_{jk\ell mn}), \quad (34)$$

$$\delta_L(l_{jk\ell mn}) = \delta_L(l_{j\ell mn}^{-1} h_{j\ell n} \triangleright' \{h_{\ell mn}, (g_{mn} g_{\ell m}) \triangleright h_{jk\ell}\} \triangleright l_{jk\ell n}^{-1} (h_{jkn} \triangleright' l_{k\ell mn}) l_{jkmn} h_{jmn} \triangleright' (g_{mn} \triangleright l_{jk\ell m})). \quad (35)$$

...all off this $\implies$

$$Z = \mathcal{N} \prod_{(jk) \in \Lambda_1} \int_G dg_{jk} \prod_{(jk\ell) \in \Lambda_2} \int_H dh_{jk\ell} \prod_{(jk\ell m) \in \Lambda_3} \int_L dl_{jk\ell m} \left(\prod_{(jk\ell) \in \Lambda_2} \delta_G(g_{jk\ell}) \right) \left(\prod_{(jk\ell m) \in \Lambda_3} \delta_H(h_{jk\ell m}) \right) \left(\prod_{(jk\ell mn) \in \Lambda_4} \delta_L(l_{jk\ell mn}) \right). \quad (36)$$

This expression can be made independent of the triangulation if one appropriately chooses the constant factor $\mathcal{N}$.

Definition

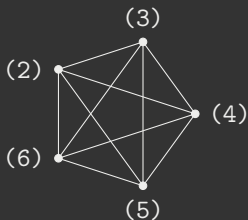
Let $\mathcal{M}_4$ be a compact and oriented combinatorial 4-manifold, and $(L \xrightarrow{\delta} H \xrightarrow{\partial} G, \triangleright, \{-, _\text{pf}\})$ be a 2-crossed module. The state sum of *topological higher gauge theory* is defined by

$$\begin{aligned}
 Z = & |G|^{-|\Lambda_0|+|\Lambda_1|-|\Lambda_2|} |H|^{|\Lambda_0|-|\Lambda_1|+|\Lambda_2|-|\Lambda_3|} |L|^{-|\Lambda_0|+|\Lambda_1|-|\Lambda_2|+|\Lambda_3|-|\Lambda_4|} \\
 & \times \left(\prod_{(jk) \in \Lambda_1} \int_G dg_{jk} \right) \left(\prod_{(jkl) \in \Lambda_2} \int_H dh_{jkl} \right) \left(\prod_{(jk\ell m) \in \Lambda_3} \int_L dl_{jk\ell m} \right) \\
 & \times \left(\prod_{(jkl) \in \Lambda_2} \delta_G(\partial(h_{jkl}) g_{kl} g_{jk} g_{jl}^{-1}) \right) \left(\prod_{(jk\ell m) \in \Lambda_3} \delta_H(\delta(l_{jk\ell m}) h_{j\ell m} (g_{\ell m} \triangleright h_{jkl}) h_{k\ell m}^{-1} h_{jkm}^{-1}) \right) \\
 & \times \left(\prod_{(jk\ell mn) \in \Lambda_4} \delta_L(l_{j\ell mn}^{-1} h_{j\ell n} \triangleright' \{h_{\ell mn}, (g_{mn} g_{\ell m}) \triangleright h_{jkl}\}_{\triangleright} l_{jk\ell n}^{-1} (h_{jkn} \triangleright' l_{k\ell mn}) l_{jkmn} h_{jmn} \triangleright' (g_{mn} \triangleright l_{jk\ell m})) \right).
 \end{aligned}
 \tag{37}$$

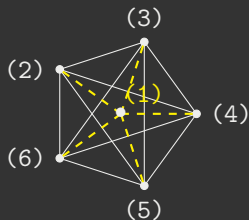
Here $|\Lambda_0|$ denotes the number of vertices, $|\Lambda_1|$ edges, $|\Lambda_2|$ triangles, $|\Lambda_3|$ tetrahedrons, and $|\Lambda_4|$ 4-simplices of the triangulation.

→ T. Radenković and M. Vojinović, [arXiv: 2201.02572](https://arxiv.org/abs/2201.02572).

>>> 1 ↔ 5 Pachner move



1 ↔ 5



	l.h.s.	r.h.s
M ₀		(1)
M ₁		(12), (13), (14), (15), (16)
M ₂		(123), (124), (125), (126), (134), (135), (136), (145), (146), (156)
M ₃		(1234), (1235), (1236), (1245), (1246), (1256), (1345), (1346), (1356), (1456)
M ₄	(23456)	(13456), (12456), (12356), (12346), (12345)

>>> 1 ↔ 5 Pachner move

	$ \Lambda_0 $	$ \Lambda_1 $	$ \Lambda_2 $	$ \Lambda_3 $	$ \Lambda_4 $
l.h.s.	5	10	10	5	1
r.h.s.	6	15	20	15	5

Right side

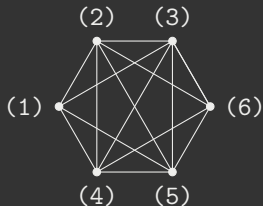
$$Z_{\text{right}}^{1 \leftrightarrow 5} = |G|^{-11} |H|^{-4} |L|^{-1} \int_{G^5} \prod_{(jk) \in M_1} dg_{jk} \int_{H^{10}} \prod_{(jk\ell) \in M_2} dh_{jk\ell} \int_{L^{10}} \prod_{(jklm) \in M_3} dl_{jklm} \cdot \left(\prod_{(jk\ell) \in M_2} \delta_G(g_{jk\ell}) \right) \left(\prod_{(jk\ell m) \in M_3} \delta_H(h_{jk\ell m}) \right) \left(\prod_{(jk\ell mn) \in M_4} \delta_L(l_{jk\ell mn}) \right) Z_{\text{remainder}} , \quad (38)$$

Left side

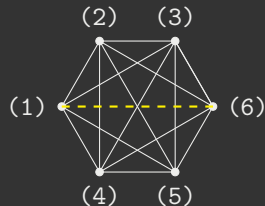
$$Z_{\text{left}}^{1 \leftrightarrow 5} = |G|^{-5} |H|^0 |L|^{-1} \delta_L(l_{23456}) Z_{\text{remainder}} . \quad (39)$$

The $Z_{\text{remainder}}$ denotes the part of the state sum that is the same on both sides of the move, and thus irrelevant for the proof of invariance.

>>> 2 ↔ 4 Pachner move



2 ↔ 4



	l.h.s.	r.h.s
M_0		
M_1		(16)
M_2		(126), (136), (146), (156)
M_3	(2345)	(1236), (1246), (1256), (1346), (1356), (1456)
M_4	(23456), (12345)	(12346), (12356), (12456), (13456)

>>> 2 ↔ 4 Pachner move

	$ \Lambda_0 $	$ \Lambda_1 $	$ \Lambda_2 $	$ \Lambda_3 $	$ \Lambda_4 $
l.h.s.	6	14	16	9	2
r.h.s.	6	15	20	14	4

Right side

$$Z_{left}^{2\leftrightarrow 4} = |G|^{-8} |H|^{-1} |L|^{-1} \int_L dl_{2345} \delta_H(h_{2345}) \left(\prod_{(jklmn) \in M_4} \delta_L(l_{jklmn}) \right) Z_{\text{remainder}} , \quad (40)$$

Left side

$$Z_{right}^{2\leftrightarrow 4} = |G|^{-11} |H|^{-3} |L|^{-1} \int_G dg_{16} \int_{H^4} dh_{126} dh_{136} dh_{146} dh_{156} \int_L dl_{1236} dl_{1246} dl_{1256} dl_{1346} dl_{1356} dl_{1456} \\ \left(\prod_{(jkl) \in M_2} \delta_G(g_{jkl}) \right) \left(\prod_{(jklm) \in M_3} \delta_H(h_{jklm}) \right) \left(\prod_{(jklmn) \in M_4} \delta_L(l_{jklmn}) \right) Z_{\text{remainder}} . \quad (41)$$

>>> Proof of $2 \leftrightarrow 4$ Pachner move invariance

- * On the left hand side of the move one has the following integrals and the integrand,

$$\int_L dl_{2345} \delta_H(h_{2345}) \delta_L(l_{23456}) \delta_L(l_{12345}). \quad (42)$$

We integrate out l_{2345} using $\delta_L(l_{12345})$. The δ -function $\delta_H(h_{2345})$ now reads,

$$\delta_H(h_{2345}) = \delta_H(e). \quad (43)$$

The remaining δ -function $\delta_L(l_{23456})$, reads

$$\begin{aligned} \delta_L(l_{23456}) = & \delta_L(l_{2456}^{-1} l_{2346}^{-1} l_{2356} (h_{256} g_{56} \triangleright h_{125}^{-1}) \triangleright' g_{56} \triangleright l_{1235} (h_{256} g_{56} \triangleright h_{125}^{-1} g_{56} \triangleright h_{135}) \triangleright' \\ & ((g_{35} \triangleright h_{123} h_{356}^{-1}) \triangleright' l_{3456}) \{g_{56} \triangleright h_{345}, (g_{56} g_{45} g_{34}) \triangleright h_{123}\}_p^{-1} (g_{56} \triangleright h_{345} (g_{56} g_{45}) \triangleright (h_{123} h_{234}^{-1}) h_{456}^{-1}) \triangleright' \\ & \{h_{456}, (g_{56} g_{45}) \triangleright h_{234}\}_p (h_{256} g_{56} \triangleright h_{125}^{-1}) \triangleright' g_{56} \triangleright l_{1345} \\ & (h_{256} g_{56} \triangleright h_{125}^{-1} g_{56} \triangleright h_{145}) \triangleright' ((g_{56} g_{45}) \triangleright l_{1234})^{-1} (h_{256} g_{56} \triangleright h_{125}^{-1}) \triangleright' g_{56} \triangleright l_{1245}^{-1}). \end{aligned} \quad (44)$$

Finally, the l.h.s. reads:

$$\boxed{l.h.s. = \delta_H(e) \delta_L(l_{23456}) = |H| \delta_L(l_{23456})}. \quad (45)$$

>>> Proof of $2 \leftrightarrow 4$ Pachner move invariance

* On the right hand side of the move there is the integral

$$\int_G dg_{16} \int_{H^4} dh_{126} dh_{136} dh_{146} dh_{156} \int_L dl_{1236} dl_{1246} dl_{1256} dl_{1346} dl_{1356} dl_{1456} \left(\prod_{(jkl) \in M_2} \delta_G(g_{jkl}) \right) \left(\prod_{(jklm) \in M_3} \delta_H(h_{jklm}) \right) \left(\prod_{(jklmn) \in M_4} \delta_L(l_{jklmn}) \right). \quad (46)$$

- * One integrates out g_{16} using $\delta_G(g_{126})$, h_{126} using $\delta_H(h_{1236})$, h_{136} using $\delta_H(h_{1346})$, and h_{146} using $\delta_H(h_{1456})$.
- * One integrates out l_{1236} using $\delta_L(l_{12346})$, l_{1246} using $\delta_L(l_{12456})$, l_{1346} using $\delta_L(l_{13456})$.
- * The remaining δ -functions on the group G reduces to $\delta_G(e)^3$,

$$\delta_G(g_{136}) = \delta_G(g_{146}) = \delta_G(g_{156}) = \delta_G(e).$$

- * One obtains that the remaining δ -functions on H reduce on $\delta_H(e)^3$,

$$\delta_H(h_{1256}) = \delta_H(h_{1356}) = \delta_H(h_{1456}) = \delta_H(e).$$

>>> Proof of $2 \leftrightarrow 4$ Pachner move invariance

* For the remaining δ -function $\delta_L(l_{12356})$,

$$\begin{aligned} \delta_L(l_{12356}) = & \delta_L(l_{2456}^{-1} l_{2346}^{-1} l_{2356} (h_{256} g_{56} \triangleright h_{125}^{-1}) \triangleright' g_{56} \triangleright l_{1235} (h_{256} g_{56} \triangleright h_{125}^{-1} g_{56} \triangleright h_{135}) \triangleright' \\ & ((g_{35} \triangleright h_{123} h_{356}^{-1}) \triangleright' l_{3456}) \{g_{56} \triangleright h_{345}, (g_{56} g_{45} g_{34}) \triangleright h_{123}\}_P^{-1} (g_{56} \triangleright h_{345} (g_{56} g_{45}) \triangleright (h_{123} h_{234}^{-1}) h_{456}^{-1}) \triangleright' \\ & \{h_{456}, (g_{56} g_{45}) \triangleright h_{234}\}_P (h_{256} g_{56} \triangleright h_{125}^{-1}) \triangleright' g_{56} \triangleright l_{1345} \\ & (h_{256} g_{56} \triangleright h_{125}^{-1} g_{56} \triangleright h_{145}) \triangleright' ((g_{56} g_{45}) \triangleright l_{1234})^{-1} (h_{256} g_{56} \triangleright h_{125}^{-1}) \triangleright' g_{56} \triangleright l_{1245}^{-1}). \end{aligned} \quad (47)$$

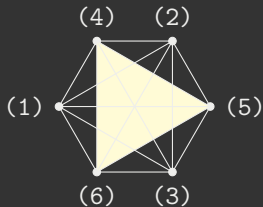
which is precisely the equation (44).

The remaining integration over the element h_{156} H and remaining integrations over the three elements l_{1246} , l_{1256} , and l_{1356} , are trivial, yielding the result of the r.h.s. to:

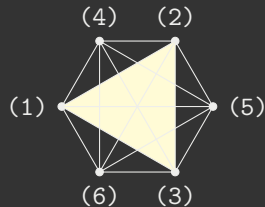
$$\boxed{r.h.s. = \delta_G(e)^3 \delta_H(e)^3 \delta_L(l_{12356}) = |G|^3 |H|^3 \delta_L(l_{12356})}. \quad (48)$$

The prefactors are $|G|^{-8} |H|^{-1} |L|^{-1}$ on the l.h.s., and $|G|^{-11} |H|^{-3} |L|^{-1}$ on the r.h.s. compensate for the left-over factors.

>>> 3 ↔ 3 Pachner move



3 ↔ 3



	l.h.s.	r.h.s
M_0		
M_1		
M_2	(456)	(123)
M_3	(1456), (2456), (3456)	(1234), (1235), (1236)
M_4	(23456), (13456), (12456)	(12356), (12346), (12345).

>>> Synopsis

- * 2-crossed modules and 3-gauge theory
- * Physically relevant models -The constrained $2BF$ actions describing the *Yang-Mills field* and *Einstein-Cartan gravity*, and constrained $3BF$ actions describing the *Klein-Gordon*, *Dirac*, *Weyl* and *Majorana fields* coupled to Yang-Mills fields and gravity in the standard way.
- * Starting from the notion of Lie 3-groups, we generalize the integral picture of gauge theory to a 3-gauge theory that involves curves, surfaces, and volumes labeled with elements of non-Abelian groups.
- * The definition of the discrete state sum model of topological higher gauge theory in dimension $d=4$.
- * We prove that the state sum is well defined, i.e., invariant under the Pachner moves and thus independent of the chosen triangulation.

>>> Synopsis

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- * Starting from the notion of Lie 3-groups, we generalize the integral picture of gauge theory to a 3-gauge theory that involves curves, surfaces, and volumes labeled with elements of non-Abelian groups.
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Thank you for your attention!